CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

November 2013

EE405 Power System Protection

Date: 21.11.2013, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 What is/are effect/s of each of following statement with respect to power system protection? **[07]**

I. If the disk of IDMT relay is made up of iron.
II. If overshoot of electro mechanical relay is zero.
III. If no spring bias is provided in directional relay.
IV. If minor earth fault in the secondary winding of transformer is not cleared by restricted earth fault protection relay.
V. If we feed |KI+V| and KI to amplitude comparator distance relay. (|KI+V| is feed to operating coil and KI is feed to restraining coil.)
VI. If the disk of IDMT relay is of circular shape.
VII. If we provide blinder characteristic instead of modified mho relay to prevent mal- operation of mho relay during power swing. (ref. figure 1.)

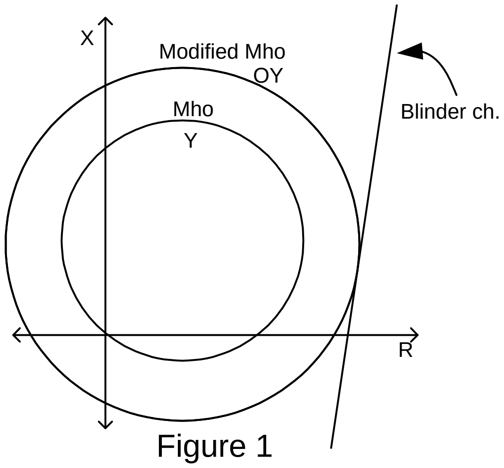

Figure 1

Q - 2 (a) Figure 2 shows a single line diagram of a power system. Relay used are directional or non-directional standard IDMT over current relays of 1 amp rating. Setting range of relay is 50-200% of rated current in step of 25%. Determine which relays required directional features and find the TMS of all relays, if TMS of R_2 and R_7 are set to 0.15. The plug settings of the relays are given as follows: **[08]**

Relays	Plug setting (% of relay rating)
R_1, R_4, R_5, R_8	100%
R_2, R_7	75%
R_3, R_6	125%

Also the high set instantaneous setting of both relays R_4 and R_5 are set to 1500% of the CT secondary rating.

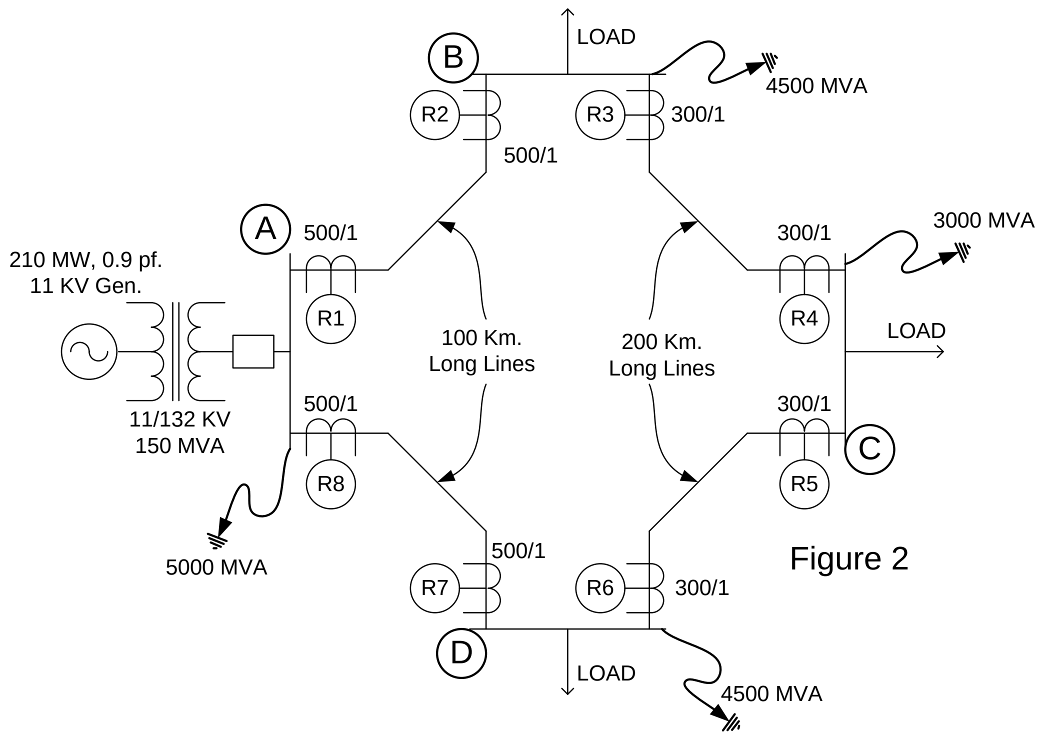

Figure 2

(b) What are the problems that distance relay encounters when a series compensated transmission line is protected by distance relay. **[04]**

(c) Magnetising inrush current of a transformer creates a problem in differential protection of a transformer. **[02]**

<u>OR</u>

Q - 2 (a) Figure 3 shows a single line diagram of a part of power system. Relay used are standard IDMT over current relays and rating of relays are as per CT secondary rating. Setting range of relay is 50-200% of rated current in step of 25%. The required details are as follows: **[08]**

Induction motor: 11 KV, 0.85 p.f., 3 phase, 2000 H.P., efficiency=90%

Circuit Breaker	Breaking Capacity (MVA)
1 and 2	2000
3 and 4	1500
5	50

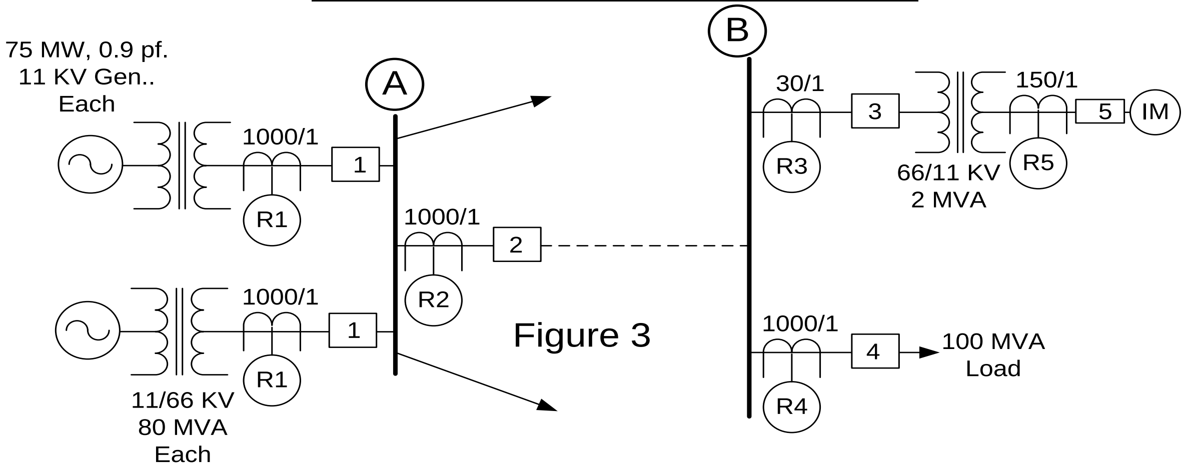

Figure 3

Determine the plug setting of all relays and TMS of R_1, R_2, R_3 if TMS of R_4 and R_5 are set to 0.1 and 0.6 respectively. Also the high set instantaneous setting of relays R_3 and R_2 are set to 1500% and 1700% of the CT secondary rating, respectively.

(b) Give Reason: For the protection of far end feeder from generator, very inverse IDMT relays are preferred over the standard inverse IDMT relay. **[03]**

(c) Explain the sentence: "The mho relay with characteristic angle 30° giving better performance for incorporating the fault resistance, proves to be worse than the mho relay with characteristic angle 70° when we compare them with respect to performance during power swing." **[03]**

Q - 3 (a) Figure 4 shows a portion of a power system network. Relay R is a mho distance relay having a MTA=60°. **[08]**

Find 2nd and 3rd zone setting of relay if

(a) Zone I covers 90% of transmission line L_2.
(b) Zone II reaches up to bus C
(c) Zone III covers 50% line length of 220 KV transmission line.
Now if bus D and E are joined together solidly, find the reach of relay for above done third zone setting.

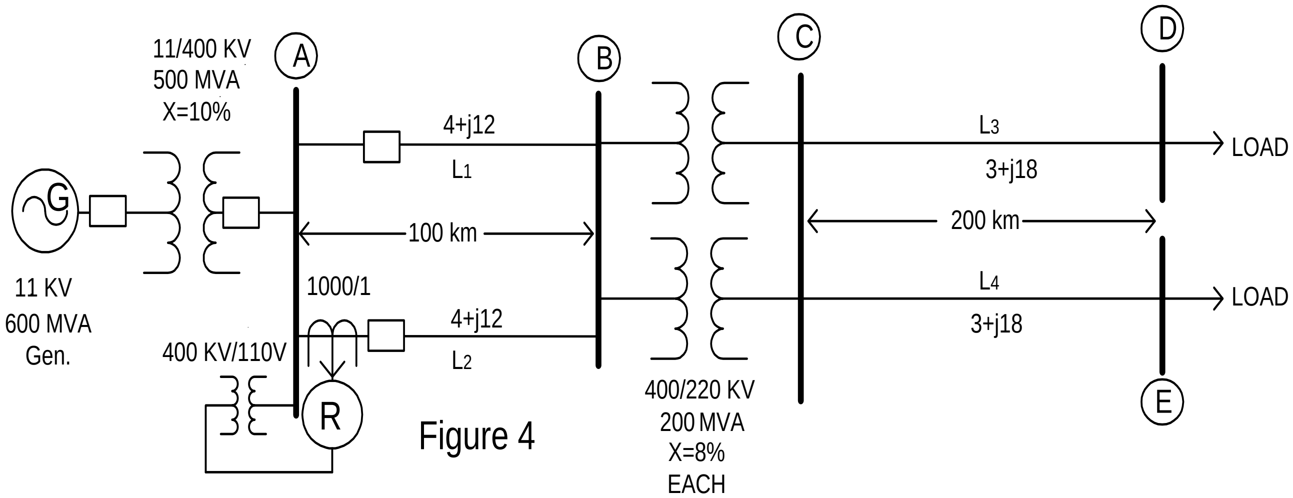

Figure 4

(b) With neat diagram explain the under reach transfer trip scheme used with carrier aided distance protection. **[06]**

OR

Q - 3 (a) A transformer is protected by biased differential relay. The relevant details as per following. **[08]**

(i) Transformer:
250 MVA, 15.75/400 KV, positive sequence impedance=10%,
Magnetizing inrush current is 8 times the rated current.
No load current is 3% of the rated current
OLTC: -7.5 % to +10% in steps of 2.5 % of rated voltage on H.V. side.
Vector Group: DY11, Rated frequency 50 Hz, Neutral solid grounded.

(ii) Current Transformers:
L.V. side : 10000/5 A star Connected
H.V. side : 300/1 A star Connected
Interposing C.T.S. connected on H.V. side
C.T.S. with Ratio Error = ± 3 %

(iii) Biased Differential Relay :
Relay Rating : 5 Amp
Basic setting: 5 % to 25 % of rating (in steps of 5 %)
Bias setting: (10 - 50 % in steps of 10 %)
H.S. inst. setting 800 to 1600% of 5Amp in step of 100%

Find out ICT ratio and draw detailed three phase wiring diagram of a transformer differential protection scheme. Also suggest suitable settings of the relay.

(b) Explain with suitable diagram: "Two over-current and one earth-fault scheme of protection does not provide adequate protection for transformer feeder." **[03]**

(c) State the following sentence is true or false. If the statement is false, correct it. **[03]**

1. First zone distance relays are never allowed to over reach.
2. One distance relay has six units. All units measure the impedance. The minimum impedance measured by any unit is always positive sequence impedance.
3. Buchholz relay is a current actuated relay.

SECTION – II

Q - 4 (a) Explain difference between Unit & Non Unit type of Protection. **[02]**

(b) Derive equation of CT over sizing Factor. **[05]**

Q - 5 (a) Drive the equation of percentage winding protected by normal E/F Protection & hence explain the concept (only concept) of protecting 100% winding of Generator stator. **[06]**

(b) Draw block diagram of Numerical relay & explain the function of each block involved. **[08]**

OR

Q - 5 (a) Explain the working of Directional comparison scheme applied to feeders connected to busbar with required diagram. **[06]**

(b) Explain Rotor Earth Fault Scheme with required diagram. **[08]**

Q - 6 (a) List out various protection applied to induction motor. **[02]**

(b) Explain Operating value Test, Reset Value Test, Operating Time Test & Reset Time Test carried out as a part of Type test for Relay. **[04]**

(c) Draw protection scheme applied to Duplicate busbar with Polarity marking on each CT. Also explain Relay Operation (Show Path of Fault Current) for any one fault assuming Bus Coupler Open. **[08]**

OR

Q - 6 (a) Explain Frame Leakage Protection for busbar. **[06]**

(b) Explain step by step method to investigate problem present in Three Phase Over Current + E/F Protection Scheme with help of Primary injection method. **[08]**

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